# F-UJI — Automated FAIR Data Assessment Tool

Tool and methodology reference for the group's FAIR assessments (software version 4.0.1, metric version v0.8)

**Version note:** the F-UJI website uses software version 4.0.1 and metric v0.8. Changes and their effects are documented on Zenodo and on GitHub.

**Disclaimer (F-UJI website):** the test results are based on preliminary data and code which still is under development; F-UJI is rapidly evolving and not yet available in a productive environment.

## What F-UJI is

F-UJI is a web-based tool that assesses the FAIRness of research data objects — Findable, Accessible, Interoperable and Reusable — against the FAIRsFAIR Metrics (FsF). It was developed in the FAIRsFAIR project ("Fostering FAIR Data Practices in Europe"), which received funding from the European Union's Horizon 2020 call H2020-INFRAEOSC-2018-2020 (grant agreement 831558). The tool is operated at f-uji.net: a user enters the persistent identifier (PID) or URL of a data object, and F-UJI harvests its metadata and runs automated metric tests.[1]

## How the assessment works

1. Input: the PID or landing-page URL of a data object (e.g. a DOI).

2. Metadata harvesting: F-UJI collects machine-readable metadata exposed by the object — DataCite JSON, schema.org / Dublin Core embedded in the landing page, content negotiation, and the files themselves.

3. Metric tests: 17 FsF metrics (FsF-F1-01MD to FsF-R1.3-02D) are evaluated; each metric is a set of automated tests with a score (earned of total) and a maturity level from 0 to 3.

4. Aggregation: per-principle (F, A, I, R) scores and FAIR levels, plus an overall FAIR level — incomplete, initial, moderate or advanced — and a percentage score.

5. Output: an interactive web report (printable to PDF) and a machine-readable JSON record of every test result.

Maturity levels per metric: 0 = incomplete, 1 = initial, 2 = moderate, 3 = advanced. The same four labels are used per principle and for the overall FAIR level.

## The 17 FAIRsFAIR metrics assessed by F-UJI

| ID | Metric |
|---|---|
| FsF-F1-01MD | Metadata and data are assigned a globally unique identifier. |
| FsF-F1-02MD | Metadata and data are assigned a persistent identifier. |
| FsF-F2-01M | Metadata includes descriptive core elements (creator, title, data identifier, publisher, publication date, summary and keywords) to support data findability. |
| FsF-F3-01M | Metadata includes the identifier of the data it describes. |
| FsF-F4-01M | Metadata is offered in such a way that it can be registered or indexed by search engines. |

| ID | Metric |
|---|---|
| FsF-A1-01M | Metadata contains access level and access conditions of the data. |
| FsF-A1-02MD | Metadata and data are retrievable by their identifier |
| FsF-A1.1-01MD | A standardized communication protocol is used to access metadata and data. |
| FsF-A1.2-01MD | Metadata and data are accessible through a standardized communication protocol which supports authentication. |
| FsF-I1-01M | Metadata is represented using a formal knowledge representation language. |
| FsF-I2-01M | Metadata uses registered semantic resources |
| FsF-I3-01M | Metadata includes qualified references between the data and its related entities. |
| FsF-R1-01M | Metadata specifies the content of the data. |
| FsF-R1.1-01M | Metadata includes license information under which data can be reused. |
| FsF-R1.2-01M | Metadata includes provenance information about data creation or generation. |
| FsF-R1.3-01M | Metadata follows a standard recommended by the target research community of the data. |
| FsF-R1.3-02D | Data is available in a file format recommended by the target research community. |

## How F-UJI was used in this project

The group ran F-UJI 4.0.1 (metrics v0.8) on 11 November 2026 against the four locations where the Formula 1 data live. The results are the basis of the four per-resource FAIR reports in this series.

| Resource | FAIR | F | A | I | R |
|---|---|---|---|---|---|
| Study of ML models for Racing LLM Appliances (racing_f1_data.zip) | **84.6%** | 85.7 | 85.7 | 66.7 | 100.0 |
| Formula 1 World Championship (1950-2020) — Kaggle dataset by Rohan Rao (rohanrao) | **84.6%** | 57.1 | 100.0 | 100.0 | 83.3 |
| zalkimanohar/Formula1-Racing-Analytics (GitHub) | **42.3%** | 42.9 | 42.9 | 33.3 | 50.0 |
| Jolpica-Ergast F1 API (Ergast-compatible endpoint) | **11.5%** | 14.3 | 28.6 | 0.0 | 0.0 |

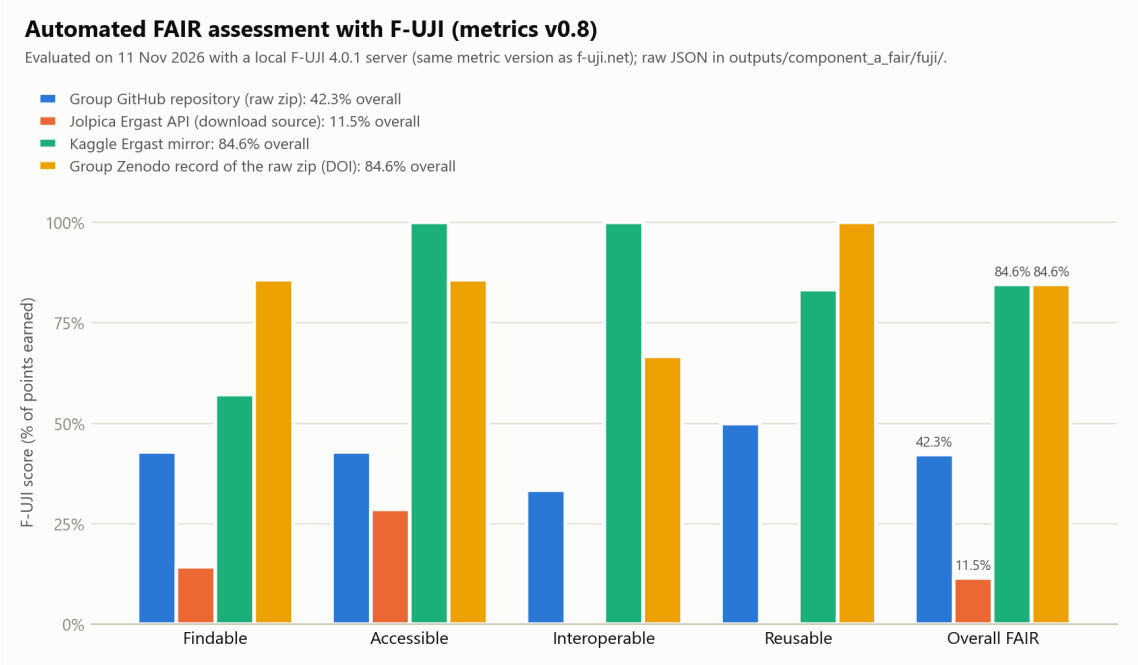



Figure A2 from the group's Component A: F-UJI scores of the four data locations.

The printed F-UJI website report of the group's Zenodo record (17 October 2026) serves as the format reference for this report series; it scored the record 84% with overall FAIR level “moderate” (Findable: advanced; Accessible, Interoperable, Reusable: moderate).

## Limitations and how they were addressed

• F-UJI checks machine-readable signals only: it scores a licence as present, not as compatible — the Kaggle mirror's CC0 claim and the Zenodo record's CC BY 4.0 both pass despite conflicting with the source's CC BY-NC-SA 4.0 terms.

• F-UJI only sees what is exposed online; it cannot read the group's local repository documentation or written evidence.

• Because of this, the group complemented the automated assessment with the RDA FAIR Data Maturity Model (all 41 indicators, scored with written evidence): overall maturity rose from 53.7% as obtained to 93.3% for the curated package, and essential indicators met went from 6/20 to 12/20.

• The tool itself is under active development (version 4.0.1, metrics v0.8); results are reproducible via the exported JSON records.

## Sources

1. F-UJI — Automated FAIR Data Assessment Tool — https://f-uji.net/index.php
2. FsF metric specification v0.8 — https://doi.org/10.5281/zenodo.15045911
3. F-UJI source code (GitHub) — https://github.com/pangaea-data-publisher/fuji
4. Group Zenodo record (sample web assessment) — https://doi.org/10.5281/zenodo.23416916
5. Group project repository — https://github.com/zalkimanohar/DataManagement